

CHAPTER 13

PROBABILITY

VERY SHORT ANSWER TYPE QUESTIONS (1 MARK)

- Find $P(A/B)$ if $P(A) = 0.4$, $P(B) = 0.8$ and $P(B/A) = 0.6$
- Find $P(A \cap B)$ if A and B are two events such that $P(A) = 0.5$, $P(B) = 0.6$ and $P(A \cup B) = 0.8$
- A soldier fires three bullets on enemy. The probability that the enemy will be killed by one bullet is 0.7. What is the probability that the enemy is still alive?
- What is the probability that a leap year has 53 Sundays?
- 20 cards are numbered 1 to 20. One card is drawn at random. What is the probability that the number on the card will be a multiple of 4?
- Three coins are tossed once. Find the probability of getting at least one head.
- The probability that a student is not a swimmer is $\frac{1}{5}$. Find the probability that out of 5 students, 4 are swimmers.
- Find $P(A/B)$, if $P(B) = 0.5$ and $P(A \cap B) = 0.32$
- A random variable X has the following probability distribution.

X	0	1	2	3	4	5
$P(X)$	$\frac{1}{15}$	k	$\frac{15k - 2}{15}$	k	$\frac{15k - 1}{15}$	$\frac{1}{15}$

Find the value of k .

10. A random variable X , taking values 0, 1, 2 has the following probability distribution for some number k .

$$P(X) = \begin{cases} k & \text{if } X = 0 \\ 2k & \text{if } X = 1 \\ 3k & \text{if } X = 2 \end{cases}, \text{ find } k.$$

SHORT ANSWER TYPE QUESTIONS (4 MARKS)

11. A problem in Mathematics is given to three students whose chances of solving it are $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{4}$. What is the probability that the problem is solved.
12. A die is rolled. If the outcome is an even number, what is the probability that it is a prime?
13. If A and B are two events such that $P(A) = \frac{1}{4}$, $P(B) = \frac{1}{2}$ and $P(A \cap B) = \frac{1}{8}$. Find $P(\text{not } A \text{ and not } B)$.
14. In a class of 25 students with roll numbers 1 to 25, a student is picked up at random to answer a question. Find the probability that the roll number of the selected student is either a multiple of 5 or of 7.
15. A can hit a target 4 times in 5 shots B three times in 4 shots and C twice in 3 shots. They fire a volley. What is the probability that atleast two shots hit.
16. Two dice are thrown once. Find the probability of getting an even number on the first die or a total of 8.
17. A and B throw a die alternatively till one of them throws a '6' and wins the game. Find their respective probabilities of winning, if A starts the game.
18. If A and B are events such that $P(A) = \frac{1}{2}$, $P(A \cup B) = \frac{3}{5}$ and $P(B) = p$ find p if events
- are mutually exclusive,
 - are independent.

19. A man takes a step forward with probability 0.4 and backward with probability 0.6. Find the probability that at the end of eleven steps he is one step away from the starting point.
20. Two cards are drawn from a pack of well shuffled 52 cards one by one with replacement. Getting an ace or a spade is considered a success. Find the probability distribution for the number of successes.
21. In a game, a man wins a rupee for a six and loses a rupee for any other number when a fair die is thrown. The man decided to throw a die thrice but to quit as and when he gets a six. Find the expected value of the amount he wins/loses.
22. Suppose that 10% of men and 5% of women have grey hair. A grey haired person is selected at random. What is the probability that the selected person is male assuming that there are 60% males and 40% females.
23. A card from a pack of 52 cards is lost. From the remaining cards of the pack, two cards are drawn. What is the probability that they both are diamonds?
24. Ten eggs are drawn successively with replacement from a lot containing 10% defective eggs. Find the probability that there is at least one defective egg.
25. Find the variance of the number obtained on a throw of an unbiased die.

LONG ANSWER TYPE QUESTIONS (6 MARKS)

26. In a hurdle race, a player has to cross 8 hurdles. The probability that he will clear a hurdle is $\frac{4}{5}$, what is the probability that he will knock down in fewer than 2 hurdles?
27. Bag A contains 4 red, 3 white and 2 black balls. Bag B contains 3 red, 2 white and 3 black balls. One ball is transferred from bag A to bag B and then a ball is drawn from bag B. The ball so drawn is found to be red. Find the probability that the transferred ball is black.
28. If a fair coin is tossed 10 times, find the probability of getting.
 - (i) exactly six heads,
 - (ii) at least six heads,
 - (iii) at most six heads.

29. A doctor is to visit a patient. From the past experience, it is known that the probabilities that he will come by train, bus, scooter by other means of transport are respectively $\frac{3}{13}$, $\frac{1}{5}$, $\frac{1}{10}$ and $\frac{2}{5}$. The probabilities that he will be late are $\frac{1}{4}$, $\frac{1}{3}$, and $\frac{1}{12}$ if he comes by train, bus and scooter respectively but if comes by other means of transport, then he will not be late. When he arrives, he is late. What is the probability that he comes by train?
30. A man is known to speak truth 3 out of 4 times. He throws a die and reports that it is six. Find the probability that it is actually a six.
31. An insurance company insured 2000 scooter drivers, 4000 car drivers and 6000 truck drivers. The probability of an accidents are 0.01, 0.03 and 0.15 respectively one of the insured persons meets with an accident. What is the probability that he is a scooter driver?
32. Two cards from a pack of 52 cards are lost. One card is drawn from the remaining cards. If drawn card is heart, find the probability that the lost cards were both hearts.
33. A box X contains 2 white and 3 red balls and a bag Y contains 4 white and 5 red balls. One ball is drawn at random from one of the bags and is found to be red. Find the probability that it was drawn from bag Y .
34. In answering a question on a multiple choice, a student either knows the answer or guesses. Let $\frac{3}{4}$ be the probability that he knows the answer and $\frac{1}{4}$ be the probability that he guesses. Assuming that a student who guesses at the answer will be incorrect with probability $\frac{1}{4}$. What is the probability that the student knows the answer, given that he answered correctly?
35. Suppose a girl throws a die. If she gets 5 or 6, she tosses a coin three times and notes the number of heads. If she gets 1, 2, 3 or 4 she tosses a coin once and notes whether a head or tail is obtained. If she obtained exactly one head. What is the probability that she throws 1, 2, 3 or 4 with the die?
36. In a bolt factory machines A , B and C manufacture 60%, 30% and 10% of the total bolts respectively, 2%, 5% and 10% of the bolts produced by

them respectively are defective. A bolt is picked up at random from the product and is found to be defective. What is the probability that it has been manufactured by machine A ?

37. Two urns A and B contain 6 black and 4 white, 4 black and 6 white balls respectively. Two balls are drawn from one of the urns. If both the balls drawn are white, find the probability that the balls are drawn from urn B .
38. Two cards are drawn from a well shuffled pack of 52 cards. Find the mean and variance for the number of face cards obtained.
39. Write the probability distribution for the number of heads obtained when three coins are tossed together. Also, find the mean and variance of the number of heads.
40. Two groups are competing for the position on the Board of Directors of a corporation. The probabilities that the first and the second groups will win are 0.6 and 0.4 respectively. Further if the first group wins, the probability of introducing a new product is 0.7 and the corresponding probability is 0.3 if the second group wins. Find the probability that the new product introduced was by the second group.

ANSWERS

- | | |
|---------------------------------|-----------------------|
| 1. 0.3 | 2. $\frac{3}{10}$ |
| 3. $(0.3)^3$ | 4. $\frac{2}{7}$ |
| 5. $\frac{1}{4}$ | 6. $\frac{7}{8}$ |
| 7. $\left(\frac{4}{5}\right)^4$ | 8. $\frac{16}{25}$ |
| 9. $k = \frac{1}{5}$ | 10. $k = \frac{1}{6}$ |
| 11. $\frac{3}{4}$ | 12. $\frac{1}{3}$ |
| 13. $\frac{3}{8}$ | 14. $\frac{8}{25}$ |

15. $\frac{5}{6}$

16. $\frac{5}{9}$

17. $\frac{6}{11}, \frac{5}{11}$

18. (i) $p = \frac{1}{10}$, (ii) $p = \frac{1}{5}$

19. 0.3678

20.

X	0	1	2
$P(X)$	$\frac{81}{169}$	$\frac{72}{169}$	$\frac{16}{169}$

21. $-\frac{91}{54}$

22. $\frac{3}{4}$

23. $\frac{1}{17}$

24. $1 - \left(\frac{9}{10}\right)^{10}$

25. $\text{var}(X) = \frac{35}{12}$

26. $\frac{12}{5} \left(\frac{4}{5}\right)^7$

27. $\frac{2}{33}$

28. (i) $\frac{105}{512}$

(ii) $\frac{193}{512}$

(iii) $\frac{53}{64}$

29. $\frac{1}{2}$

30. $\frac{3}{8}$

31. $\frac{1}{52}$

32. $\frac{22}{425}$

33. $\frac{25}{52}$

34. $\frac{12}{13}$

35. $\frac{8}{11}$

36. $\frac{12}{37}$

37. $\frac{5}{7}$

38. Mean = $\frac{8}{13}$, Variance = $\frac{1200}{287}$

39.

X	0	1	2	3
P(X)	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

Mean = $\frac{3}{2}$

Variance = $\frac{3}{4}$

40. $\frac{2}{9}$